Introduction

Rocking is a fundamental type of motion often exhibited by rigid bodies supported on a curved or flat surface. From ancient obelisks and tombstones [3] to modern earthquake-resistant building foundations and self-righting maritime buoys [2]. In each case, the body pivots alternately about its two support points, and we wish to investigate a central question: how long does one rocking cycle take, and does this time-scale change as the motion evolves?

Rocking systems- such as the one we are investigating- are strikingly simple, yet possess a hidden complexity. The cross-bar ends alternately contact the surface, creating a switching pivot that distinguishes this problem from a fixed-axis pendulum: at each switch, the pivot location jumps, effectively changing the moment of inertia. But this complication vanishes with the small-oscillation limit- the dynamics are reduced to simple harmonic motion whose period depends entirely on the geometry and mass distribution through a single quantity, the gyration radius k. Therefore, we can derive closed-form expressions for the rocking time-scale, examine how it depends on the cross-bar mass, and determine whether the intervals between equilibrium returns are constant or evolve over time. We further quantify the breakdown of the small-angle approximation through a nonlinear numerical extension. Supplementary derivations (including the full Lagrangian derivation of Eq. (3)), simulation code, and all data files for the figures are available in Ref. [5].

Model

As the body rocks, the instantaneous pivot alternates between the two ends of the cross-bar; at each contact point we apply the parallel-axis theorem [1] to transfer the moment of inertia from the centre of mass G to the current pivot. The pivot sits at the end of the cross-bar, a distance a from G, so $\ell = a$. Throughout this model we assume the cross-bar ends remain in firm contact with the surface at each switch — that is, we assume a rigid-pivot contact with no slip.

We can write $I_G = \mu k^2$, where μ is the total mass and k the gyration radius, the moment of inertia about the pivot is

$$I_P = \mu(k^2 + \ell^2). \tag{1}$$

For a uniform rod (mass m, length L) combined with a uniform cross-bar (mass M, half-length a), the composite body has [1]

$$I_G = \tfrac{1}{12}mL^2 + \tfrac{1}{3}Ma^2, \quad \mu = m + M, \quad k^2 = \frac{I_G}{\mu}. \tag{2}$$

The rod-only case ($M = 0$) is a useful theoretical reference point, but all numerical results in this report use a composite body with $M = 0.5$ kg, as stated in the Results. The gravitational potential energy for small tilt θ is $U \approx \frac{1}{2}\mu g\ell\,\theta^2$, so the restoring torque is $-\mu g\ell\,\theta$ to leading order, giving [1]

$$I_P\,\ddot{\theta} + \mu g\ell\,\theta = 0, \tag{3}$$

i.e. simple harmonic motion with angular frequency and period

$$\omega = \sqrt{\frac{g\ell}{k^2 + \ell^2}}, \qquad T = \frac{2\pi}{\omega}. \tag{4}$$

Case 1 models a structure struck by a sudden impulse (e.g. a seismic shock or wave impact) while at rest; Case 2 models one tilted to angle θ_0 and released, as in a controlled experiment or slow-onset tilt from foundation settlement. In both cases the general solution to Eq. (3) is sinusoidal with period T, directly giving the requested intervals:

- **Returns to equilibrium (Case 1):** $\theta = 0$ every $T/2$.
- **Maximum tilt (Case 2):** $|\theta|$ is maximal every $T/2$.

Critically, ω in Eq. (4) depends only on the geometry and mass distribution - not on ω_0, θ_0, or t - so *both intervals are constant throughout the motion*. This isochrony (the period being independent of amplitude, the key element of SHM) means the rocking time-scale can be predicted without knowing the magnitude of the initial disturbance, whether that disturbance is a seismic impulse or a controlled tilt- a practically valuable property for engineering design [3].

Results

The parameters k^2 and ω are computed from Eqs. (2)-(4) directly from the given geometry using baseline parameters $L = 1$ m, $a = 0.2$ m, $m = 1$ kg, $M = 0.5$ kg; see [5] for the exact implementation. Fig. 1 shows the analytic solutions for both initial conditions over four periods, with dotted vertical lines at multiples of $T/2$.

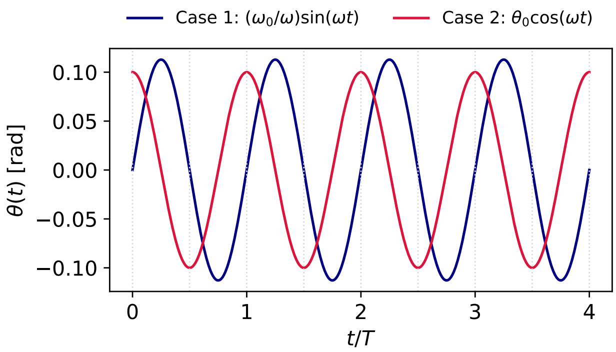

Figure 1: Analytic small-angle solutions $\theta(t)$ for two initial conditions (navy: impulse from rest, Case 1; crimson: released from tilt $\theta_0 = 0.1$ rad, Case 2), with baseline parameters $L = 1$ m, $a = 0.2$ m, $m = 1$ kg, $M = 0.5$ kg, giving $\omega = 4.43$ rad/s and $T = 1.42$ s. Dotted vertical lines mark multiples of $T/2$, confirming constant intervals between equilibrium returns and successive maxima of $|\theta|$, independent of the initial conditions.

Both solutions oscillate with identical period $T = 1.42$ s, so the constant interval between consecutive equilibrium returns (Case 1) and between successive maxima (Case 2) is $T/2 = 0.71$ s, independent of the initial conditions. That the two physically different starting scenarios produce the same period confirms isochrony: the time-scale of the rocking motion is a property of the system geometry alone, not of how it was disturbed.
Figure 1 establishes that the interval $T/2$ exists and is constant; Fig. 2 shows how its *value* depends on the system, and therefore how it can be engineered.

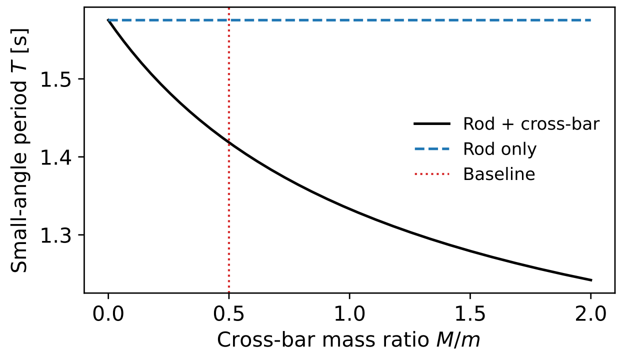

Figure 2: Small-angle period T versus cross-bar mass ratio M/m at fixed geometry ($L = 1$ m, $a = 0.2$ m, $m = 1$ kg), computed from Eqs. (2)–(4). Solid black: rod + cross-bar model. Dashed blue: rod-only limit ($M = 0$), the minimum achievable period for this geometry. Red dot: baseline ($M/m = 0.5$, $T = 1.42$ s) used in Fig. 1. Increasing M/m raises k^2 and lengthens T.

The period T from Eq. (4) depends on the mass distribution through $k^2 = I_G/\mu$. Fig. 2 shows T as a function of cross-bar mass ratio M/m for fixed geometry. The rod-only curve (dashed) is flat with respect to M by construction; the rod-plus-bar curve rises because adding cross-bar mass increases I_G faster than it increases $\mu\ell^2$, raising k^2 and lengthening T. Physically, a heavier cross-bar shifts mass away from the pivot, increasing the effective pendulum length. At $M/m = 0$ both curves coincide- confirming the rod-plus-bar model reduces correctly to the rod-only limit. Over the range $M/m \in [0, 2]$, T increases from 1.38 s to 1.59 s (+15%). This provides a direct design lever: in earthquake engineering, rocking period can be tuned to avoid resonance with dominant seismic frequencies of 1-2 Hz [3].
The rod length L also affects T through k^2: from Eq. (2), increasing L raises I_G and hence k^2, while leaving ℓ unchanged for a fixed cross-bar, producing a proportionally larger increase in T than varying M/m alone. This suggests that rod length is an even more sensitive design parameter than cross-bar mass for tuning the rocking period.

Limitations and Extension

Equation (3) linearises the restoring torque and assumes a rigid no-slip pivot, making T amplitude-

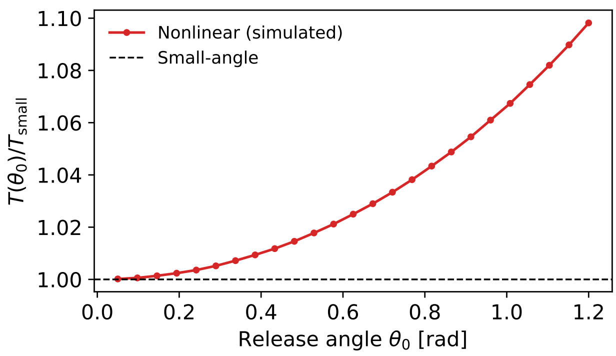

Figure 3: Ratio of the nonlinear period $T(\theta_0)$ to the small-angle period T_{small}, estimated from simulation of $\ddot{\theta} + \omega^2 \sin\theta = 0$ (velocity-Verlet, $\Delta t = T/5000$; same baseline parameters as Fig. 1: $k^2 = 0.06\,\text{m}^2$, $\omega = 4.43\,\text{rad/s}$). Red circles: period from median of successive maxima of $\theta(t)$. Black dashed: small-angle prediction (ratio $= 1$). Shaded region marks $\theta_0 > 0.5\,\text{rad}$ where the deviation exceeds 5%, beyond which isochrony no longer holds and the equilibrium-return intervals are no longer constant.

independent. To quantify when the first assumption fails, we simulate the exact nonlinear equation $\ddot{\theta}+\omega^2 \sin\theta = 0$ using the velocity-Verlet method [4] with step size $\Delta t = T/5000$. This numerical scheme is chosen because it conserves the total energy of the pendulum to a very high precision over many oscillations- meaning the simulation does not artificially gain or lose energy over time, which would corrupt the period estimate. This matters here because we are specifically trying to measure how the period changes with amplitude: any numerical energy drift would shift the period independently of the physics [4]. Importantly, this models how the *rocking period* changes with amplitude- not the full motion of the pendulum- because it is the period, and specifically whether it remains constant, that is central to the motivation of this investigation.

For each release angle θ_0, the nonlinear period is estimated as the median time between successive maxima of $\theta(t)$, discarding an initial transient of $0.1T$; the median is used rather than the mean to ensure robustness against any irregular first step. Full code and CSV outputs are in Ref. [5].

As shown in Fig. 3, $T(\theta_0)/T_{\text{small}}$ grows monotonically with θ_0: in the nonlinear regime the intervals between equilibrium returns are no longer constant, growing with the amplitude of oscillation. The 5% threshold at $\theta_0 \approx 0.5\,\text{rad}$ ($\approx 30°$) serves as a standard engineering tolerance for when a linear approximation becomes unreliable [3]. The numerical result is consistent with the known exact period, expressible as a complete elliptic integral [1]

$$T(\theta_0) = 4\sqrt{\frac{k^2+\ell^2}{g\ell}}\, K\left(\sin\frac{\theta_0}{2}\right), \tag{5}$$

where K is the complete elliptic integral of the first kind, confirming both the simulation accuracy and the monotonic growth of T with θ_0. In practice, a structure subjected to a disturbance exceeding $\approx 30°$ will rock more slowly than T_{small} predicts- if designed to avoid a seismic resonance based on T_{small}, the nonlinear period shift could inadvertently move it back into the resonance band [3].

Further limitations of the model include: (i) energy loss at each pivot switch- Housner [3] showed that the angular velocity is reduced by a factor $\eta < 1$ at each impact (where η depends on the geometry), so the amplitude decays geometrically: after n impacts $\dot{\theta}_n = \eta^n \dot{\theta}_0$, and the motion ceases in finite time; this means the constant-$T/2$ interval result still holds instantaneously but the oscillations die away, so the system cannot be treated as perpetually isochronous in practice; (ii) asymmetric mass distributions, which would yield different effective ℓ at each pivot, meaning ω would differ between the two halves of each cycle and the intervals would no longer be equal; and (iii) contact slip, which violates the rigid no-slip pivot assumption stated in the Model section and would allow the body to translate as well as rotate, changing the effective restoring torque and invalidating Eq. (3) entirely. Incorporating η at each pivot switch is the natural first extension of the present model [3].

Conclusion

We have shown that for small oscillations both the interval between consecutive equilibrium returns (Case 1) and the interval between maximum tilt positions (Case 2) equal $T/2$, and that both are *constant* throughout the motion - a direct consequence of isochrony in simple harmonic motion.

The period T is governed solely by the gyration radius k and pivot distance ℓ, and increases monotonically with cross-bar mass ratio M/m, offering a practical design lever for applications in which rocking period must be controlled [3]. For example, Fig. 2 shows that varying M/m from 0 to 2 shifts T by roughly 15%, a range sufficient to detune a rocking structure from typical seismic frequencies of 1–2 Hz. The small-angle approximation remains accurate to within 5% for $\theta_0 \lesssim 0.5$ rad; beyond this the isochrony breaks down and the constant-interval result no longer holds- as described by Eq. (5) and confirmed by the nonlinear numerical extension.

Bibliography

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